FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 — SUB-STRAND 4.1: STATISTICS I	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS
This is your Final Explanation task. Over the past 9 lessons, you have been investigating the "Maize Harvest Mystery of Nakuru County." A KALRO officer visited 20 smallholder farmers and collected harvest data — but the raw list was chaotic and impossible to use. Your job now is to show what you have learned by writing a complete, well-organised explanation that answers the driving question: "How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action?" Read each section prompt carefully. Use the exemplar responses as a guide for the quality and depth expected. Support every claim you make with data, calculations, or diagrams where asked. Write clearly — imagine your audience is a Nakuru County government official who needs to trust your analysis to make a real fertilizer-allocation decision.

Section 1: Collecting and Organising Raw Data	
Look back at the raw list of 20 maize harvest values (in 90-kg bags) from the Nakuru County farmers. Explain why the raw list alone is not useful for decision-making. Then describe TWO methods you would use to organise this data, and explain what each method reveals that the raw list does not. Use specific examples from the Nakuru data in your answer.	When the KALRO officer first wrote the 20 harvest values on the chalkboard in the order she collected them, the data was overwhelming. There was no pattern visible — a value of 4 bags sat next to a value of 27 bags, making it impossible to tell at a glance what a 'typical' harvest looked like. Raw data is collected data that has not yet been sorted, grouped, or summarised, and on its own it cannot answer questions about trends or typical values. To make sense of this data, two powerful organising methods are a tally-and-frequency table and ordering the data from smallest to largest (an ordered array). A frequency table groups the harvest values into class intervals — for example, 1–5 bags, 6–10 bags, 11–15 bags, and so on — and records how many farmers fall into each group. This immediately shows you where most farmers are clustered. For the Nakuru data, if 8 out of 20 farmers fall in the 11–20 bag range, the frequency table makes that visible at a glance, whereas the raw list hides it.

	An ordered array simply re-lists all 20 values from the smallest (4 bags) to the largest (31 bags). This makes it easy to find the middle value and to spot whether there are extreme values (outliers) that might distort a calculated average. For example, once ordered, you can immediately see that one farmer's unusually high yield of 31 bags stands far apart from the rest — a detail lost in the chaotic raw list. Both methods transform confusing raw data into organised information that tells a story, which is the first step toward helping the county government make a fair decision.

Section 2: Representing Data Visually	
Choose TWO types of graphs or charts that are appropriate for representing the Nakuru maize harvest data. For each graph: (a) state its name and describe how it is constructed, (b) explain what pattern or story it reveals about the farmers' harvests, and (c) explain why that visual representation is more useful to a county government official than a list of numbers. You may sketch or describe a graph in your answer.	Two graphs that are well suited to the Nakuru maize harvest data are a frequency histogram and a bar graph of grouped data. A frequency histogram is constructed by placing the class intervals (e.g., 1–5, 6–10, 11–15, 16–20, 21–25, 26–31 bags) on the horizontal axis and the frequency (number of farmers) on the vertical axis. Bars are drawn for each interval with no gaps between them, because the data is continuous. For the Nakuru data, the histogram would show a peak in the middle intervals — suggesting that most farmers harvested between 11 and 20 bags — with the bars becoming shorter toward the extremes. This shape (roughly a mound) tells the county official immediately that a harvest of around 11–20 bags is the most common outcome, not the unusually low value of 4 bags or the unusually high value of 31 bags. A bar graph can be used to compare harvests across separate categories — for example, if the 20 farmers came from five different sub-locations in Nakuru County (such as Njoro, Molo, Rongai, Subukia, and Gilgil), a bar graph with one bar per sub-location and bar height representing the total or average yield per sub-location would allow the official to compare areas side by side. This is far more powerful than reading a list, because the eye can immediately see which sub-location is underperforming and may need more fertilizer. Both graphs convert numbers into shapes and comparisons that the human eye processes quickly. A county official reviewing 20 numbers would need several minutes of careful reading; a well-drawn histogram communicates the same story in seconds, making data-driven decisions faster and more reliable.

Section 3: Calculating and Interpreting Measures of Central Tendency	
Using the Nakuru County maize harvest data, calculate the mean, median, and mode. Show your working clearly. Then answer this question: The county government wants ONE number to represent the 'typical' harvest. Which measure of central tendency would you recommend, and why? In your answer, explain a	Using a sample set of the 20 Nakuru farmer harvest values (in 90-kg bags): 4, 7, 8, 9, 10, 11, 12, 12, 14, 15, 15, 15, 17, 18, 19, 20, 22, 24, 27, 31. Mean: Add all values and divide by 20. Sum = 4+7+8+9+10+11+12+12+14+15+15+15+17+18+19+20+22+24+27+31 = 299 Mean = 299 ÷ 20 = 14.95 bags ≈ 15 bags. Median: Because there are 20 values (an even number), the median is the average of the 10th and 11th values in the ordered list. 10th value = 15, 11th value = 15

situation where each measure could be misleading.	Median = (15 + 15) ÷ 2 = 15 bags. Mode: The value that appears most often is 15 bags (it appears 3 times). Mode = 15 bags. In this particular dataset, all three measures give 15 bags, which builds confidence. However, in real data they often differ, and the choice matters enormously. I would recommend the median as the most reliable single representative number for the county government. Here is why: the median is the middle value — exactly half of all farmers harvested more than it and half harvested less. It is not pulled up or down by extreme values (outliers). If one exceptionally lucky farmer near Molo harvested 80 bags due to irrigation, the mean would jump sharply upward and suggest a 'typical' harvest far above what most farmers actually achieved — leading the government to allocate too little fertilizer. The median would barely change. The mean can be misleading when there are outliers, as shown above. The mode can be misleading when the most common value is at one extreme of the data — for example, if most farmers had a very poor year and the mode was 4 bags, using the mode might cause the government to underestimate typical productivity. Choosing the right measure is not just a mathematical decision — it is a matter of fairness to Nakuru's farming communities.

Section 4: Connecting Your Analysis to the Anchoring Phenomenon	
Return to the moment in the video where the KALRO officer asks: 'Which single number best represents all our farmers — and how do we even begin to make sense of all this data?' Using everything you have learned across all 9 lessons, write a direct, evidence-based answer to her question. Explain how the tools of statistics (data collection, organisation, graphs, and measures of central tendency) work together as a system to turn raw data into a decision-making tool. Refer specifically to the Nakuru County context.	Officer, the answer to your question has two parts: the right tool depends on your data, and no single tool works alone — they must be used as a system. Step 1 — Organise first. Before any calculation is possible, the 20 harvest values need to be organised. Sorting them into an ordered array and building a frequency table grouped by class intervals (1–5, 6–10, etc.) immediately transforms the chaotic chalkboard list into a readable summary. This step is not optional — it is the foundation. Step 2 — Visualise to find the story. A frequency histogram of the Nakuru data reveals that the majority of farmers harvest between 11 and 20 bags. This pattern is completely invisible in the raw list. A bar graph comparing sub-locations within Nakuru (Njoro, Molo, Rongai, Subukia, Gilgil) can show the county government exactly which areas have the lowest yields and therefore the greatest need for subsidised fertilizer. Step 3 — Calculate the right average. For the Nakuru data, the median (15 bags) is the most trustworthy single representative number, because it sits at the true centre of the data and is not distorted by the unusually high harvest of 31 bags reported by one farmer. If the government used the mean without checking for outliers, they might overestimate typical productivity and under-allocate fertilizer to struggling farmers. Step 4 — Make the decision. With a well-organised dataset, a clear histogram, and a reliable median of 15 bags per farmer, the county government can now set a baseline: any farmer consistently harvesting below 15 bags per season is below the county typical and should be prioritised for fertilizer subsidies.

	Statistics did not change the farmers' yields — but it gave the county government the clarity to act fairly and effectively. That is the power of turning raw data into organised information.

Section 5: Reflection — Your Role as a Data Citizen	
Think beyond Nakuru County. Describe ONE other real situation in your own community — in Nairobi, Kisumu, Mombasa, Kericho, or wherever you live — where collecting, organising, and representing data could help a decision-maker take more effective action. Identify: (a) what data would be collected and how, (b) which graph would best represent it and why, and (c) which measure of central tendency would be most appropriate and why. Show that you understand statistics as a tool for community empowerment, not just a set of calculations.	In Kisumu County, school feeding programmes provide lunch — often githeri or uji — to primary school pupils. A county nutrition officer wants to know whether children are receiving adequate portions and whether attendance improves on days when food is provided. This is a real decision-making challenge where statistics can help. (a) Data collection: The officer would visit a sample of 30 schools across Kisumu sub-counties (Kisumu Central, Seme, Muhoroni, etc.) and record, for each school, the average daily attendance (as a percentage) on feeding days versus non-feeding days. She would use a structured data collection form, recording the school name, sub-county, total enrolment, and attendance figures for 10 consecutive school days. This gives organised, reliable raw data rather than guesses. (b) Best graph: A double bar graph would be most appropriate. For each sub-county, one bar would represent average attendance on feeding days and a second bar would represent average attendance on non-feeding days. Placing these bars side by side for each sub-county makes the comparison immediate and visual. A decision-maker can see at a glance whether the feeding programme is associated with higher attendance — and whether the effect is stronger in some sub-counties than others (perhaps rural Muhoroni vs. urban Kisumu Central). (c) Best measure of central tendency: The mean attendance percentage would be most appropriate here, because attendance data recorded as percentages is unlikely to have extreme outliers — most schools will cluster between 60% and 95%. The mean gives the officer a reliable overall picture. If one school had an unusually low attendance due to a local flood event (not related to the feeding programme), the officer should note and exclude that value before calculating, showing responsible use of statistics. Statistics gives every student the power to be a data citizen — someone who can look at a community problem, collect evidence, organise it honestly, and present findings that lead to better decisions for real people.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Data Organisation and Representation	Clearly explains why raw data is unhelpful and accurately describes two or more organising methods (e.g., frequency table, ordered array) and two appropriate graphs (e.g., histogram, bar graph) with correct construction details and specific reference to the Nakuru data. Explains what each method reveals.	Explains why raw data is limited and describes at least one organising method and one graph correctly, with some reference to the Nakuru context. Minor gaps in explaining what each method reveals.	Attempts to describe data organisation or graphs but explanations are vague, incomplete, or contain errors. Limited or no connection to the Nakuru data.

Calculation of Measures of Central Tendency	Correctly calculates mean, median, and mode with full working shown. Accurately identifies which measure best represents the Nakuru data, provides a well-reasoned justification, and explains a specific situation where each measure could be misleading.	Correctly calculates at least two of the three measures with working shown. Recommends a measure with a reasonable justification but may not fully explain when each measure can mislead.	Attempts calculations but makes errors in two or more measures, or shows little or no working. Recommendation of a measure is unsupported or absent.
Connection to the Anchoring Phenomenon	Provides a direct, evidence-based answer to the KALRO officer's question. Clearly explains how data organisation, graphs, and measures of central tendency work together as a system. All reasoning is grounded in specific Nakuru County details.	Answers the officer's question and references two or more statistical tools. Connection to the Nakuru context is present but may be general rather than specific.	Attempts to answer the officer's question but relies on vague or general statements. Little or no specific connection to the Nakuru phenomenon or the tools studied.
Application to a New Community Context	Identifies a clear, relevant real-world community scenario with a Kenyan context. Correctly justifies the chosen graph and measure of central tendency for that new context with specific reasoning. Demonstrates understanding of statistics as a tool for community empowerment.	Identifies a relevant scenario and makes reasonable choices of graph and measure with partial justification. Kenyan context is present. Some understanding of statistics beyond mere calculation is shown.	Scenario is vague or not community-relevant. Graph or measure choices are missing or unjustified. Little evidence that statistics is understood as a decision-making tool.
Clarity, Structure, and Use of Mathematical Language	Writing is well organised across all five sections. Mathematical vocabulary (mean, median, mode, frequency, outlier, class interval, histogram, etc.) is used accurately and consistently. Calculations are clearly laid out and easy to follow.	Writing is mostly organised and mathematical vocabulary is used correctly in most places. Calculations are present but may lack full clarity or layout. Minor language errors do not obscure meaning.	Writing is disorganised or incomplete in multiple sections. Mathematical vocabulary is missing, misused, or inconsistent. Calculations are difficult to follow or absent.